

GhostRes Edge Brief Project Description

Snapdragon AI Lab Build and Present Challenge 2026

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GhostRes Edge is a privacy focused desktop application for restoring low resolution video locally. It uses Real ESRGAN neural super resolution to reconstruct edges, textures, and fine detail that ordinary interpolation cannot recover. The product is designed around a modular inference boundary so that the current CUDA desktop engine and a future Snapdragon QNN engine can share one user workflow.

Problem

Legacy clips, compressed recordings, and low resolution camera footage are difficult to reuse because enlarged pixels become soft and blocky. Cloud restoration services may require a user to upload personal video and usually hide performance tradeoffs. Creators need a clear local workflow that lets them choose quality or speed while retaining control of their media.

Product

The GhostRes Edge desktop interface lets a user select a video, choose Fast 2x or Quality 4x enhancement, start restoration, and open the completed output. The application reports output resolution, frames processed, model inference latency, AI frames per second, and total processing time. This makes the tradeoff visible instead of presenting AI restoration as a black box.

Architecture

Layer	Role	Current implementation
Desktop workflow	Video selection, scale choice, progress, and output access	CustomTkinter application
Video pipeline	Read frames, validate output dimensions, write restored video, record metrics	OpenCV and VideoResult telemetry
Inference boundary	Keep enhancement backends swappable without changing the application workflow	EnhancementEngine interface
Desktop backend	Run neural super resolution on the development laptop	CUDA Real ESRGAN x2plus and x4plus with tiled inference
Snapdragon validation path	Prepare a portable tile model for Qualcomm acceleration	ONNX export and Qualcomm AI Hub QNN NPU profiling

Technical Implementation

GhostRes Edge builds on RRDBNet and Real ESRGAN models. The current local engine supports x2plus and x4plus model weights, 128 pixel tiling, and 10 pixel tile padding to fit the RTX 3060 development GPU. The app is intentionally separated from model code: the UI calls a common enhancement interface while the engine owns model loading and frame inference. A fixed shape ONNX export of the x4 model accepts float32 image tiles shaped [1, 3, 128, 128] and produces [1, 3, 512, 512] outputs.

Measured Evidence

The following results are measured at different levels of the project and are not presented as directly comparable. The desktop result measures end to end video processing on the CUDA development machine. The Snapdragon result measures one exported fixed size inference tile on Qualcomm hosted hardware.

Environment	Workload	Measured result
RTX 3060 desktop app	95 frame 864 by 480 video using Fast 2x	1728 by 960 output in 165.46 seconds 1720 ms AI latency per frame 0.58 AI FPS
Qualcomm AI Hub Snapdragon X Elite CRD	Real ESRGAN x4 ONNX tile float32 [1, 3, 128, 128] QNN NPU	65.4 ms inference latency 0.2 MB peak memory

Snapdragon Readiness

The Real ESRGAN x4 tile was exported to ONNX, validated with the ONNX checker, compiled, and profiled through Qualcomm AI Hub on a Snapdragon X Elite CRD running Windows 11 using the QNN NPU compute unit. This validates the model path on Snapdragon hardware. The current desktop demonstration itself runs on the RTX CUDA development machine; it is not represented as a completed native Snapdragon desktop application. The next engineering step is a QNN or ONNX Runtime QNN EnhancementEngine that runs the compiled model on a Snapdragon PC while retaining the same UI and video pipeline.

What Makes It Different

GhostRes Edge is not only a model demo. It combines a usable local desktop workflow, switchable quality and speed modes, measured inference telemetry, and a deliberate route from a proven CUDA prototype to Snapdragon QNN deployment. It also makes its limits explicit: super resolution generates plausible visual detail rather than proving missing detail was present in the source, and the current intermediate video writer does not yet preserve audio.

Next Steps

The immediate roadmap is to add a QNN backed inference engine, benchmark full video pipeline performance on a physical Snapdragon PC, preserve source audio during output creation, and package the desktop workflow with explicit model and runtime setup. The architecture already isolates these changes to the engine and output layers so the user experience remains stable.